Phase Shifted Full Bridge (PSFB) Converter- I

Introduction

In the realm of power electronics, efficient and controlled energy transfer is paramount, especially in applications like charging infrastructure where isolation between input and output is often required for safety and performance. This lecture introduces the Phase-Shifted Full Bridge (PSFB) converter, a widely utilized isolated DC-DC converter topology. Understanding its operation is crucial for designing robust and high-performance power conversion systems, particularly those requiring soft-switching capabilities to minimize losses and improve efficiency.

The PSFB converter builds upon the foundational knowledge of full-bridge converters and their various Pulse Width Modulation (PWM) strategies. While traditional PWM methods involve varying the turn-on duration of switches, the PSFB converter employs a unique phase-shift modulation technique. This approach allows for output voltage regulation by controlling the phase difference between the switching legs, offering distinct advantages in achieving Zero Voltage Switching (ZVS) for enhanced efficiency and reduced electromagnetic interference (EMI). This lecture delves into the operational modes of the PSFB converter, highlighting the mechanisms that enable soft switching and the critical conditions required for its successful implementation.

Core Concepts

Term	Definition	Significance
`Phase-Shifted Full Bridge (PSFB) Converter`	An isolated DC-DC converter where output voltage is regulated by varying the phase shift between the switching signals of the two half-bridges.	Enables soft-switching (ZVS) for improved efficiency and reduced switching losses, crucial for high-power applications.
`PWM Type 1`	A PWM strategy for full-bridge converters where diagonal switches turn off simultaneously.	Often struggles to achieve Zero Voltage Switching (ZVS) during turn-on for all switches, leading to higher switching losses.

Term	Definition	Significance
`PWM Type 2`	A PWM strategy for full-bridge converters where diagonal switches turn off at different times (staggered).	Offers a higher possibility of achieving ZVS, especially for the leading leg, making it more efficient than Type 1.
`Zero Voltage Switching (ZVS)`	A soft-switching technique where a power switch turns on when the voltage across it is zero.	Significantly reduces turn-on switching losses, improves converter efficiency, and lowers EMI.
`Leading Leg`	One of the two half-bridges in a full-bridge converter, typically the one whose switches initiate the phase shift.	In PSFB, the leading leg switches are more likely to achieve ZVS turn-on due to the energy stored in the output inductor.
`Lagging Leg`	The other half-bridge in a full-bridge converter, whose switching is delayed relative to the leading leg.	ZVS for the lagging leg can be more challenging to achieve, depending on load conditions and parasitic elements.
`Dead Time`	A short, intentional delay between the turn-off of one switch and the turn-on of its complementary switch in a half-bridge.	Essential for allowing parasitic capacitances to charge/discharge, enabling ZVS, and preventing shoot-through.
`Leakage Inductance (L_lkg)`	The portion of the transformer's primary inductance that does not couple to the secondary winding.	Plays a crucial role in the resonant transitions during PSFB operation, particularly in achieving ZVS for the lagging leg.
`Output Capacitance (C_oss)`	The parasitic capacitance across the drain-source terminals of a MOSFET switch.	Utilized in soft-switching transitions; its charging and discharging facilitate

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		ZVS by shaping voltage waveforms.
`Phase Shift Modulation`	A control method where the output voltage is controlled by varying the phase difference between the gate signals of the leading and lagging legs.	Allows for continuous output voltage regulation while maintaining fixed switch turn-on times (typically 50% duty cycle).

Detailed Analysis

1. Evolution from Traditional PWM to Phase-Shifted Control

The lecture begins by revisiting conventional PWM strategies for full-bridge converters, categorizing them into two main types based on the turn-off behavior of diagonal switches. This sets the stage for understanding the advantages of phase-shifted modulation.

- **PWM Type 1**: In this strategy, diagonal switches (e.g., S1 and S4, or S2 and S3) turn off simultaneously. While simple, this method typically does not facilitate Zero Voltage Switching (ZVS) during turn-on for all four switches, leading to higher switching losses.
- **PWM Type 2**: Here, diagonal switches turn off at different, staggered times. This approach significantly improves the chances of achieving ZVS, particularly for the "leading leg" switches. The "lagging leg" switches may or may not achieve ZVS depending on factors like leakage inductance energy and primary current. The output capacitance of the switches generally aids in ZVS turn-off for all switches.
- **Phase Shift Modulation (PSM)**: Unlike traditional PWM where the turn-on period (duty ratio) of switches is varied, PSM keeps the turn-on time of all switches (S1, S2, S3, S4) fixed, typically at half the switching period (Ts/2). Output voltage regulation is achieved by introducing a controlled phase shift (delta) between the leading leg and the lagging leg. Increasing this phase shift reduces the effective duty cycle seen by the transformer, thereby lowering the output voltage. This method inherently aligns with PWM Type 2 characteristics, making it suitable for ZVS.

2. PSFB Operating Modes and ZVS for the Leading Leg

The operation of the PSFB converter is analyzed through distinct modes, focusing on the current paths and voltage transitions that enable soft switching. The analysis assumes ideal switches initially, then incorporates non-idealities like output capacitances and leakage inductance.

- **Mode-0 (t < t0): Power Delivery Phase**
- Switches S1 and S4 are conducting, applying the input voltage (Vin) across the primary side of the transformer (VAB = Vin).
- The primary current (ip) is a reflection of the output inductor current (IL), which is assumed to be nearly constant due to a large output inductor.
- The voltage across the secondary rectifier (Vr) is NVin (where N is the transformer turns ratio Ns/Np), causing the output inductor current to rise with a positive slope (nVin - V0)/L.
- **Mode-1 (t0 < t < t1): Leading Leg Transition (S1 Turn-off)**
- At t = t0, the gate pulse to S1 is removed, turning it off.
- The primary current (ip), which was flowing through S1, now charges the output capacitance of S1 (C1) and discharges the output capacitance of S2 (C2).
- VC1 increases from 0, and VC2 decreases from Vin.
- The voltage VAB (across the primary) starts falling from Vin towards 0, with a slope determined by the primary current (I1) and the sum of the output capacitances (C1 + C2).
- During this period, S4 remains on, and the secondary side rectifier diodes DR1 and DR2 become forward biased, effectively short-circuiting the secondary winding. This causes Vr to become zero, and the output inductor current starts falling with a slope of -V0/L.
- This transition continues until VC2 reaches 0 and VC1 reaches Vin.
- **Mode-2 (t1 < t < t2): Leading Leg ZVS Turn-on (S2)**
- At t = t1, VC2 reaches 0, and the body diode of S2 (D2) turns on, providing a path for the primary current.
- Since the voltage across S2 is now effectively zero (due to D2 conduction), the gate pulse for S2 can be applied, turning it on with Zero Voltage Switching (ZVS).
- The time duration for this transition (t1 - t0), denoted as T01, must be sufficient for VC2 to discharge to zero. This time is approximately 2C*Vin / I1, where C is the output capacitance and I1 is the primary current.

- **Dead Time Condition for Leading Leg ZVS**: To ensure ZVS for S2, the dead time (the delay between S1 turning off and S2 turning on) must be greater than T01 (2CVin / I1). This guarantees that D2 conducts before S2 is gated on, ensuring zero voltage across S2 at turn-on. The primary current I1 can be approximated as NI0 (reflected output current).
- During this mode, S2 and S4 are on, and VAB is zero. The secondary side remains short-circuited, and Vr is zero.
- **Mode-3 (t2 < t < t3): Lagging Leg Transition (S4 Turn-off)**
- At t = t2, the gate pulse to S4 is removed, turning it off. S2 remains on.
- The primary current (ip) now charges the output capacitance of S4 (C4) and discharges the output capacitance of S3 (C3).
- VC3 decreases from Vin, and VC4 increases from 0.
- The voltage VAB starts to become negative (VAB = -VC4).
- Crucially, during this transition, the leakage inductance (L_lkg) of the transformer comes into play. The primary current, which is no longer a direct reflection of the output inductor current, resonates with the output capacitances C3 and C4.
- The secondary side diodes DR1 and DR2 remain forward biased, keeping Vr at zero. This effectively short-circuits the secondary, causing the entire VAB voltage to be imposed across the leakage inductance.
- The equivalent circuit for this mode shows L_lkg in resonance with C3 and C4 (effectively 2C if C3=C4=C). The primary current (ip) and capacitor voltages (VC3, VC4) undergo sinusoidal changes.
- The conditions for ZVS turn-on of S3 (the complementary switch to S4 in the lagging leg) are more complex and depend on the energy stored in the leakage inductance and the primary current at the instant of S4 turn-off. These conditions will be discussed in detail in the subsequent lecture.

Key Takeaways

- [ ] Phase-shifted full-bridge converters regulate output voltage by varying the phase shift between switching legs, not by changing individual switch duty cycles.
- [ ] The turn-on time for all switches in a PSFB converter is typically fixed at Ts/2.
- [ ] PSFB converters inherently operate similarly to PWM Type 2 strategies, which are more conducive to achieving Zero Voltage Switching (ZVS).

- [ ] ZVS turn-on for the leading leg switches (e.g., S2 after S1 turns off) is achieved by utilizing the primary current to discharge the switch's output capacitance to zero before turn-on.
- [ ] A critical "dead time" is required between the turn-off of one switch and the turn-on of its complementary switch to allow for ZVS.
- [ ] For the leading leg, the dead time must be greater than 2CVin / (NI0) to ensure ZVS turn-on.
- [ ] During the lagging leg transition, the transformer's leakage inductance resonates with the switch output capacitances, complicating ZVS conditions.
- [ ] When the secondary side is short-circuited by the rectifier diodes, the voltage across the transformer primary (VAB) is effectively applied across the leakage inductance.

Conclusion

The Phase-Shifted Full Bridge converter represents a significant advancement in isolated DC-DC power conversion, primarily due to its inherent ability to achieve Zero Voltage Switching (ZVS). By meticulously controlling the phase shift between the leading and lagging legs, designers can regulate the output voltage while simultaneously minimizing switching losses, a critical factor for improving efficiency and reducing thermal stress in high-power applications. The detailed analysis of its operating modes reveals how parasitic elements, such as switch output capacitances and transformer leakage inductance, are strategically utilized to facilitate soft transitions.

While the leading leg's ZVS conditions are relatively straightforward to meet by ensuring adequate dead time, the lagging leg presents a more complex challenge due to the resonant interaction between leakage inductance and switch capacitances. This foundational understanding of PSFB operation, particularly the mechanisms for leading leg ZVS, is essential for anyone involved in the design and optimization of modern power converters. The subsequent discussions will further elaborate on the intricacies of lagging leg ZVS and other practical considerations, building upon the principles established in this lecture to provide a comprehensive view of this powerful converter topology.